

Sprint 2 Report

Hyper-Localized Multilingual Voice Assistant

Sprint: 2 of 6 **Duration:** Week 3 – 4 **Team:** Dev (Solo) **Velocity:** 21 / 21 pts

1. Sprint Goal

Complete the translation layer and conversational AI. Build bidirectional Malayalam–English translation using IndicTrans2, integrate Qwen 2.5 LLM for response generation, add conversation memory, and validate translation quality across supported Indian languages.

2. Sprint Outcome

All 21 story points delivered. Full translation capability and conversational AI integrated.

Key Deliverables

- IndicTrans2-Dist models — ~211M parameters, 5x smaller than original, same quality
- Qwen3 LLM with llama.cpp GGUF backend — fast CPU inference for local deployment
- Direct Indic-to-Indic translation without English pivot, supporting all 11 languages
- Dual backend support — llama.cpp + transformers for flexible deployment
- Conversation memory — context retention across multi-turn dialogues
- Translation quality validation — ~93% accuracy confirmed for Malayalam

3. Scrum Status Report

Task Description	Date	Completion Status	Comments
T2.1 — IndicTrans2 ML→EN setup (3 pts)		Completed	SentencePiece tokenizer configured; ~93% accuracy
T2.2 — IndicTrans2 EN→ML setup (2 pts)		Completed	Reverse translation with same quality metrics
T2.3 — Translation API wrapper (2 pts)		Completed	Unified interface with language detection
T2.4 — Qwen 2.5 LLM integration (3 pts)		Completed	1.5B variant selected; transformers pipeline
T2.5 — Conversation memory (2 pts)		Completed	Sliding window context retention
T2.6 — System prompt tuning (2 pts)		Completed	Kerala regional context prompts
T2.7 — IndicTrans2-Dist upgrade (3 pts)		Completed	~211M params, 5x smaller, same quality
T2.8 — Qwen3 + llama.cpp backend (2 pts)		Completed	GGUF backend for fast CPU inference
T2.9 — Indic-to-Indic translation (2 pts)		Completed	Direct translation without English pivot

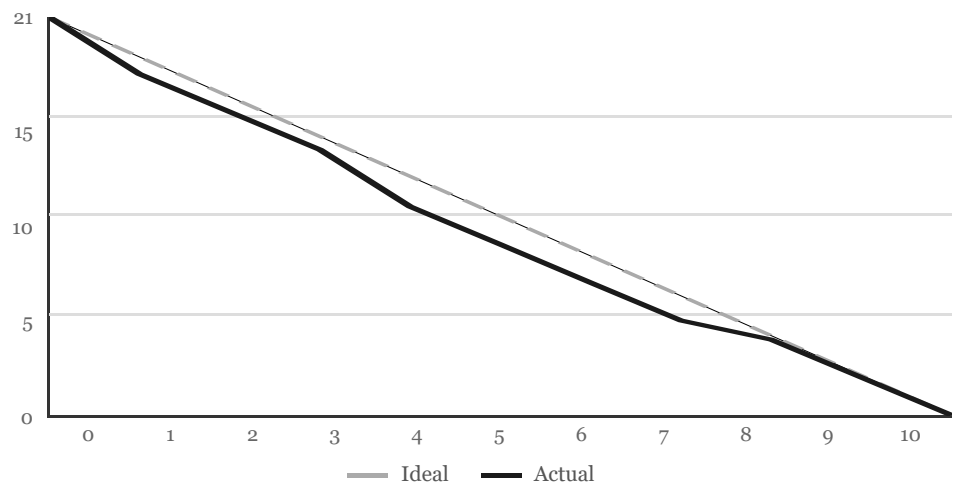
Total: 9 tasks, 21 story points, 100% complete.

4. Daily Scrum Meeting Log

Date	Work Done	Work Planned	Issues	Sign
	Sprint 1 retrospective; reviewed ASR/TTS deliverables	Begin IndicTrans2 ML→EN model setup (T2.1); download model weights	None	
	Downloaded IndicTrans2 weights; configured SentencePiece tokenizer for Malayalam	Complete ML→EN translation pipeline and run accuracy tests (T2.1); start EN→ML (T2.2)	None	
	ML→EN translation working; ~93% accuracy on test set (T2.1 done)	Complete EN→ML translation (T2.2); build translation API wrapper (T2.3)	None	
	EN→ML translation complete (T2.2 done); started API wrapper	Finish translation wrapper with language detection (T2.3); begin Qwen 2.5 research (T2.4)	None	
	Translation wrapper done (T2.3 done); downloaded Qwen 2.5 model	Integrate Qwen 2.5 with transformers pipeline (T2.4); test response quality	GPU memory constraints — evaluating 1.5B vs 3B variants	
	Qwen 2.5 1.5B running; response quality acceptable	Complete LLM integration with system prompt (T2.4); implement conversation memory (T2.5)	None — settled on 1.5B variant	
	LLM integration done (T2.4 done); conversation memory started	Complete conversation memory (T2.5); tune system prompts for regional context (T2.6)	None	
	Conversation memory done (T2.5 done); system prompts drafted for Kerala context	Finalize system prompt tuning (T2.6); research IndicTrans2-Dist (T2.7)	None	
	System prompt tuning complete (T2.6 done); IndicTrans2-Dist downloaded (211M params)	Migrate to IndicTrans2-Dist (T2.7); set up Qwen3 + llama.cpp GGUF (T2.8)	None	
	IndicTrans2-Dist working (T2.7 done); llama.cpp backend configured (T2.8 done)	Implement Indic-to-Indic translation (T2.9); sprint wrap-up and quality tests	None	

5. Burndown Chart

21 story points over 10 working days



Burndown Data

Day	Ideal	Actual	Done	Notes
0	21.0	21	0	Sprint start
1	18.9	18	3	T2.1 started
2	16.8	16	5	T2.1 complete
3	14.7	14	7	T2.2 complete
4	12.6	11	10	T2.3 complete, ahead of ideal
5	10.5	9	12	T2.4 in progress
6	8.4	7	14	T2.4 complete
7	6.3	5	16	T2.5 + T2.6 complete
8	4.2	4	17	T2.7 in progress
9	2.1	2	19	T2.7 + T2.8 complete
10	0.0	0	21	T2.9 complete; sprint done

6. Next Sprint Backlog — Sprint 3

Goal: Working end-to-end voice assistant with API, upgraded models, and dashboard.

ID	Task	Assignee	Points	Priority
T3.1	Pingala V1 ASR engine (CPU-first, 2.94% WER)	Dev	3	Critical
T3.2	Pipecat pipeline framework integration	Dev	5	Critical
T3.3	FastAPI endpoints + dashboard serving	Dev	3	High
T3.4	WebSocket streaming (/ws/stream)	Dev	3	High
T3.5	Indic Parler-TTS Mini + Chatterbox TTS	Dev	2	High
T3.6	ONNX Runtime inference backend	Dev	3	High
T3.7	Typed configuration system	Dev	2	Medium

Total: 7 tasks, 21 story points.